Practice workbook · Explain stereoscopic vision and how it helps judge distance

What stereoscopic vision is, why two eyes judge distance better than one, and an investigation comparing catching with one and both eyes.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: HBI-EXPLAIN-STEREOSCOPIC-VISION

1. Question 1 · part b · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b only.

1 (a) The box lists parts of the eye and parts of the ear.

auditory nerve ciliary body cochlea cornea
eardrum optic nerve ossicles
pupil retina semicircular canals

The table has descriptions of parts of the eye and ear.

Complete the table by matching the descriptions, using parts from the box.

(7)

Description	Part
contains light sensitive cells	
vibrates when receives sound waves	
helps the body balance	
controls the amount of light entering the eye	
transfers vibrations across the middle ear	
turns vibrations into nerve impulses	
refracts light entering the eye	

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1. Question 1 · part b · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b only.

(b) Explain why a person using both eyes finds it easier to catch a ball than a person who is blind in one eye.

(3)

(Total for Question 1 = 10 marks)

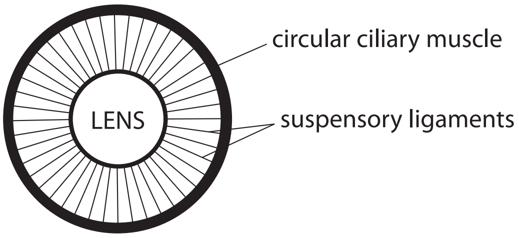
Working and answer

2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	**Focusing on a distant object**
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(c) Design an investigation to show that it is easier for a person to catch a ball if they have stereoscopic vision.

Your answer should include experimental details and be written in complete sentences.

(6)

(Total for Question 5 = 12 marks)

2. Question 5 · part b · 3 marks · Working and answer

Working and answer

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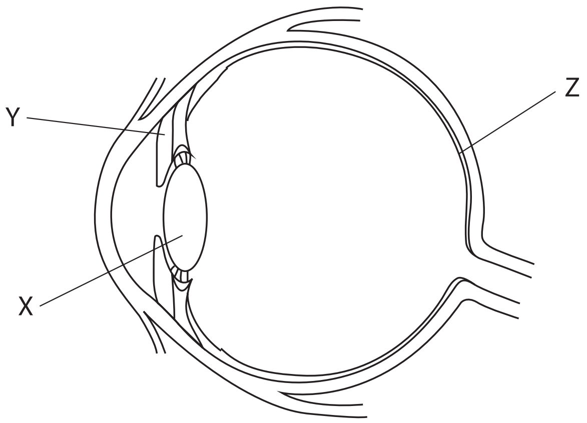
3. Question 1 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure

- structure X ●
- structure Y ●
- structure Z ●

Function

- ● transmits electrical impulses to the brain
- ● refracts light rays
- ● protects the eye from pathogens
- ● detects light rays
- ● controls the shape of the lens
- ● controls the size of the pupil

3. Question 1 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

☒ **A** being able to judge depth
☒ **B** being able to see in colour
☒ **C** having a wide field of view
☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.

Healthy eye
Eye with cataract

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(i) Describe the cause of cataracts.

(2)

....

....

....

(ii) State one treatment for cataracts.

(1)

....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

....

(Total for Question 1 = 10 marks)

3. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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